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Explore the entrepreneurial process of AI start-ups from the perspective of opportunity

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Abstract

Digital technologies, such as artificial intelligence (AI), big data, and the Internet of Things, are becoming increasingly mature. They have become the driving force behind a new wave of innovation and entrepreneurship activities worldwide. AI is the core of the contemporary digital technology industry. New high technologies bring greater opportunities for entrepreneurs to carry out entrepreneurial activities and bring unprecedented challenges to traditional entrepreneurial activities and theories. The existing literature on the impact of AI on entrepreneurial activities is still in the exploration stage. Based on this, this study aims to explore the entrepreneurial process of AI start-ups, combining with existing literature, select AI unicorn start-ups ByteDance for a case study, and apply the grounded theory method to analyse and summarise the entrepreneurial process of Internet start-ups with AI. Thus, it can provide insights for many AI start-ups and contribute to the existing entrepreneurial research.

KEYWORDS

AI, entrepreneurial opportunity, entrepreneurial process

1 | INTRODUCTION

China has seen a boom in artificial intelligence (AI) business start-up activity in recent years. On the one hand, the total number of Chinese AI companies is the second largest in the world. By 2021, Chinese AI companies have exceeded 2205. The number of AI unicorn start-ups also ranks among the top in the world. Among the 32 unicorns in the world, 10 in China are slightly lower than the 17 in the United States. On the other hand, the development of China's AI industry is mainly driven by application demand. Computer vision, language, natural language processing, and others are hot areas for the development of AI companies, and a number of excellent start-ups have emerged. China's AI industry has initially formed an industrial ecosystem with basic technical support, AI technology, and AI applications, of which the number of AI companies in the application side accounts for 84.05% of the highest. In addition, China's AI industry has huge market potential. AI technology has a wide range of application scenarios in new economic fields, new business models and new lifestyles, and the transformation and upgrading of traditional industries. According to IiMedia Research 2020 estimates, China's AI output value will be close to $400 billion in 2025, making it a new blue ocean for entrepreneurial activities. However, the success rate of AI start-ups in practice is less than 5%, and more and more AI companies will fail in the future. The entrepreneurial process, such as the development of the opportunity by the entrepreneur, is an important factor in the success or failure of the venture. Technologies such as AI and big data can affect entrepreneurs' perception, improve the efficiency of entrepreneurial opportunity development,

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and influence the creation of new ventures (Cai et al., 2019). Therefore, we need to understand the process behind AI business start-ups to provide references and guidance for more AI business start-ups.

There are two main types of understanding of AI in entrepreneurship in the academic community. One view is that AI entrepreneurship is an entrepreneurial activity in the technical application field of AI (Wu, 2016), that is, there are two types of entrepreneurship: "AI + industry" and "industry + AI". AI is mainly based on various elements such as machine learning, AI planning, natural language processing, and knowledge representation. These elements that are related to AI technologies are very useful for entrepreneurial activities (Khalid, 2020). The development of AI is based on the continuous development of new technologies. New technologies also affect entrepreneurial activities, so the use of AI has a very important role in carrying out entrepreneurial activities. The development of AI technology has advanced rapidly, but in the short term, AI cannot reach the level of human intelligence, and modern entrepreneurial activities can only use AI as a tool for human services, and there are still a lot of innovative entrepreneurial opportunities in the application of AI technology (Zheng, 2017). Another view is that AI plays the role of an "intelligent human" and thus empowers the process of entrepreneurial activity, using AI as a new entrepreneurial tool (Liu & Wang, 2020). They argue that AI entrepreneurship is a way for entrepreneurs to collaborate with AI to develop entrepreneurial opportunities actively, so it is essentially a way to empower the entire process of traditional entrepreneurial activities with the help of AI technology. It is also a new "invention approach" that can reshape the innovation and entrepreneurship process (Cockburn et al., 2018). At present, "industry + AI" has become a hot spot of entrepreneurial practice, and the empowerment of AI for various industries has appeared everywhere. This study focusses on the entrepreneurial process of AI firms.

Opportunity is one of the important paths to studying the entrepreneurial process of start-ups. Entrepreneurship scholars focus on new ventures' creation and value creation process by cutting through entrepreneurial opportunities (Bansal et al., 2021; Chen & Lei, 2021; Hou et al., 2018; Yeşilkaya & Yıldız, 2021). They have used entrepreneurial opportunities as a link to reveal the static entrepreneurial process from different perspectives such as resources, environment, and cognition (Ao & Liu, 2014; Liu et al., 2019). In recent years, the academic community has focussed on studying the dynamics of the entrepreneurial process, which has driven entrepreneurship research to explore more on the continuity of the entrepreneurial process at the level of theory construction. The dynamism and complexity of the entrepreneurial process come from the interaction among entrepreneurs and the uncertain environment. Developing entrepreneurial opportunities determines that there is no fixed procedure for entrepreneurial activities to refer to deal with the uncertainty (Zhang, 2018). In AI business start-up activities, the rapid update of new technologies and the rapid changes in consumers' lifestyles and consumption patterns expose entrepreneurs to a highly uncertain environment for creating new businesses. The relevance, development, and generation of AI and other digital technologies are different from traditional technologies (Li et al., 2021), which changes a series of related activities in which entrepreneurs take advantage of new opportunities to create value and influences the entrepreneurial process (Cai et al., 2019). AI technology has an important impact on developing entrepreneurial opportunities from an opportunity perspective. The continued advancement of AI technology enables entrepreneurs to identify more entrepreneurial opportunities (Brown, 2017; Davidsson et al., 2017) and makes it easier for entrepreneurs to identify and exploit entrepreneurial opportunities more easily (Agrawal et al., 2017; Liu et al., 2020). AI technology has changed the industrial development model, innovation approach, and entrepreneurial process (Du & Liu, 2019). Although a small number of studies have focussed on the AI enterprise process in terms of entrepreneurial opportunities, and scholars have attempted to establish a research framework for the study of the entrepreneurial process of AI enterprises (Cai et al., 2019; Liu et al., 2020), the study of the AI entrepreneurial enterprise process remains at the stage of theoretical exploration, and there is a lack of case or empirical studies examining the dynamic process of AI enterprise entrepreneurship.

Therefore, the study of the dynamic process of AI start-ups from the perspective of entrepreneurial opportunities can fill the gap in entrepreneurship research in AI technology and play an important role in guiding the practice of AI technology innovation and entrepreneurship in China.

2 | THEORETICAL BACKGROUND

2.1 | A theoretical model for the entrepreneurial process of AI enterprises

Entrepreneurial opportunity is the starting point for conducting entrepreneurship and is the core of entrepreneurial activity (Khan et al., 2022; Shane & Venkataraman, 2000). According to the opportunity school of entrepreneurship research, entrepreneurship is

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essentially the process of identifying, evaluating, and developing opportunities around entrepreneurial opportunities. Entrepreneurial opportunities are situations in which new "means-ends" relationships are used to create new products and services or introduce new modes of production (Eckhardt & Shane, 2003). The opportunity discovery view suggests that entrepreneurial opportunities exist objectively in the marketplace. Entrepreneurs can search for information about the market or technological changes in the external environment to identify opportunities to provide new products or services (Dutta & Crossan, 2010). The opportunity creation view considers that entrepreneurs subjectively construct opportunities, who perceive changes in the external environment and continuously explore and create new opportunities through various entrepreneurial means (Alvarez & Barney, 2007). There is a difference in the level of study between the opportunity discovery view and the creation view; opportunity discovery usually tends to study entrepreneurs observing the external environment at the microlevel (e.g., individuals and group pairs), whilst opportunity construction mostly observes the world at the organisational or industry level (Si et al., 2016).

The entrepreneurial process has been the focus of entrepreneurship research, and entrepreneurial opportunity is the center of entrepreneurial process research (Luo & Bai, 2021). There are some classic models of the entrepreneurial process, such as the process model consisting of people, opportunity, environment, resources, and transactional behavior (Sahlman, 1999), and the matching of the three elements of the team, resources, and opportunity constituting the dynamic process of entrepreneurship (Timmons & Spinelli, 2004). Bruyat and Julien (2001) constructed the dynamic process theory of value creation between entrepreneurs and new enterprises, as shown in Figure 1. According to the theory, the general entrepreneurial process is a process of close interaction between entrepreneur and the new firm under the action of the external environment. That is, in the entrepreneurial process, the value of the new venture is not only realised but also the knowledge, skills, and social networks of the entrepreneur are changed. The process of entrepreneurship and new venture interaction is necessarily accompanied by the pursuit of entrepreneurial opportunities, in which entrepreneurial opportunities are also closely related to time (Vaghely & Venkataraman, 2010). The model emphasises the dynamic equilibrium between the entrepreneur and the creation of a new venture in the entrepreneurial process and the influence of the external environment. The analysis of the regularity of the entrepreneurial process in terms of entrepreneurial opportunities and the variability of the entrepreneurial process in terms of the resource endowment of entrepreneurs is highly consistent with the direction of research on the dynamics and complexity of the entrepreneurial process (Zhang, 2018). The rapid development and innovation of other new technologies such as AI has made the environment for business creation and growth increasingly uncertain. Entrepreneurs with rich knowledge, skills, and social networks can grasp the changing trends of the external environment, such as new technologies, seize new opportunities in time, complete developing and utilizing new opportunities, and thus realise new value. Bruyat's entrepreneurial process model (Bruyat & Julien, 2001) provides a solid theoretical foundation for this study, so this study uses it as a basis to further analyse the dynamic entrepreneurial process of AI firms.

As the main initiators of entrepreneurial activities, entrepreneurs play a very important role in the entrepreneurial process, and entrepreneurial traits are the key factors that affect the success or failure of entrepreneurship (Xue, 2018). Entrepreneurial traits, such as human capital, are important influencing factors in identifying, discovering, and developing entrepreneurship (Wang & Cai, 2011). They have an important impact on the whole

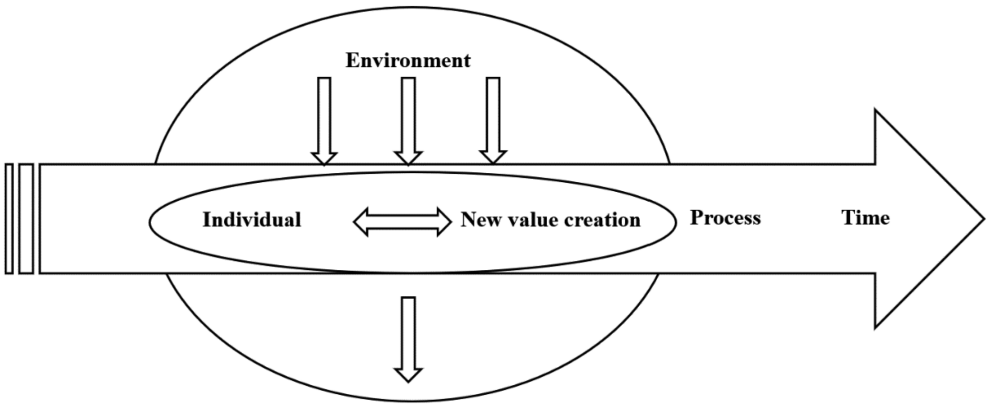


FIGURE 1 Bruyat and Julien model

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entrepreneurial process. The individual and social traits of entrepreneurs can influence the growth, performance, and process of entrepreneurial ventures and are related to the success of the venture (Timmons & Spinelli, 2004). Individual traits of entrepreneurs include psychological traits such as the need for achievement, responsibility, self-efficacy, and behavioral traits such as learning, alertness, and motivation. In contrast, social traits are mainly reflected in social capital (social connections), prior industry, management, and entrepreneurial experience. Although many scholars have studied the influence of entrepreneurial traits, behaviors, cognition, and social capital on the creation, growth, and performance of new ventures (Wang, 2016), the dynamic interaction process between entrepreneurs and new venture creation has been ignored.

The essence of entrepreneurship research is to uncover the "black box" of new value-creating activities (Lin & Jiang, 2005), and entrepreneurial opportunities have always been the core of entrepreneurial process research. Existing research has explored the entrepreneurial process, but in the emerging era of information intelligence, changes in new technologies such as AI have significantly altered the traditional entrepreneurial process. Therefore, it is of wide practical and theoretical significance to reveal the "black box" of the entrepreneurial process in the context of digital technologies such as AI.

3 | RESEARCH METHODOLOGY

3.1 | Method

Based on the relevant AI entrepreneurship literature review, this study focusses on exploring the entrepreneurial process of AI start-ups from the perspective of entrepreneurial opportunities. Case studies are reliable research methods for constructing and verifying new theories that provide reasonable explanations for new phenomena in real practice. The application of AI technology is inseparable from algorithms, computing power, data, and scenarios. Although AI technology has been widely used in most industries, AI technology is well suited for the development of the Internet industry in terms of the current speed and breadth of application and the benefits produced, and current domestic and foreign Internet start-ups have used AI technology to establish a competitive advantage in the market and gain significant development.

In the Chinese Internet industry, ByteDance ranks as the world's number one AI unicorn start-up, with a valuation of more than $140 billion. ByteDance, founded in 2012, was the first technology company to apply AI to the mobile Internet scene. Since its establishment, it has launched many personalised information content products such as Toutiao and Douyin and built an APP factory in the intelligent era by relying on the advantages of AI.

3.2 | Data collection

The main source of research information in this study is secondary collected, and rich and effective secondary information can be used as a source of information for the case study. In the process of collecting information on ByteDance, we read and repeatedly screened all the information on the enterprise, and according to the principle of triangulation verification, this study cross-compared and verified the information obtained from several different channels and eliminated the information that did not conform to triangulation verification to ensure the reliability and validity of this study. The secondary information collected from the companies mainly includes (1) video materials of public interviews or speeches of the company's founder Zhang Yiming and his executive team Zhang Nan; (2) relevant media reviews, journal columns, or magazines published on the Internet; (3) literature or reviews about the companies from China National Knowledge Infrastructure and other databases; and (4) relevant reports and industry information obtained from the official websites of the companies. The above data collection provided strong information support to obtain reliable research conclusions for this study.

In order to deeply explore the entrepreneurial process of AI-enabled start-ups, this study utilises the analytic method of rooting theory to code the original data level by level. Each sub-category is first coded at the first level, and the main category is refined through the second level coding, and then the corresponding analytical framework is constructed. The data analysis process keeps a comparative analysis idea all the time and constructs a model of the entrepreneurial process.

4 | ANALYSIS AND CONCLUSION

In the era of information intelligence, Internet industry start-ups are rapidly developing under different situations from traditional entrepreneurship, such as reduced constraints of key resources, blurred entrepreneurial boundaries, a highly uncertain entrepreneurial environment, and difficulty predetermining entrepreneurial goals (Liu & Wang, 2020). The entrepreneurial process presents

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the characteristics of fragmentation, interactivity, dynamism, and iterative innovation without boundaries (Yu et al., 2018), especially in the AI technology-empowered Internet content-based start-ups. In this paper, based on the classical entrepreneurship model of Bruyat-Julien model, we first analyse the traits of ByteDance entrepreneurs and the path of ByteDance creation at the opportunity level, respectively, and then analyse the entrepreneurial process of AI technology-enabled Internet start-ups from the perspective of entrepreneurial opportunity. From the perspective of opportunity, the entrepreneurial process of such companies mainly shows two characteristics: on the one hand, entrepreneurs generate certain creative ideas in interaction with external AI technology, Internet industry, and resource environment, and combine them with their existing "means" (technical ability, a priori knowledge, and personal network) to build a novel entrepreneurial opportunities. On the other hand, the company takes traffic as the core of development, based on the characteristics of infinite expansion and continuous regeneration of data traffic itself (Liu et al., 2020), and the characteristics of openness, borderlessness, and strong interaction of entrepreneurial activities empowered by AI technology (Yu et al., 2018), and creates a "traffic construction, expansion, and monetisation" development model, and creates more new entrepreneurial opportunities by using the massive data flow.

4.1 | Characteristics of entrepreneurs

Entrepreneurs influence the establishment, survival, and development of start-ups. In addition to seizing the opportunities of the era of mobile Internet and AI, the most important thing is that the "leader" of the company accurately grasps the industry opportunities. Through the analysis of relevant information (Table 1), we found that, on the one hand, the entrepreneurs of ByteDance already had rich AI technology, entrepreneurial experience, and social network when they started the company, and these foundations have accumulated the essential conditions for their entrepreneurial activities. The rich knowledge, industry, management, and entrepreneurial experience accumulated by the entrepreneurs due to their experience in software development and continuous entrepreneurship in the Internet industry help them to improve their ability to identify entrepreneurial opportunities (Eckhardt & Shane, 2003), better understand market needs (Politis, 2010), and more easily grasp the Internet industry trends. On the other hand, the entrepreneur's own excellent attributes, such as alertness to opportunities, innovation, risk-taking, and a strong need for achievement, are also important factors in ensuring the success of their ventures (Shane & Venkataraman, 2000). Entrepreneurs remain innovative, risk-taking, and have strong achievement needs in an uncertain environment of technological development and demand markets that drives them to start multiple businesses. Alertness makes entrepreneurs keenly aware of the need for personalised information portals as the information explosion brought by mobile Internet inconveniences users in sifting through information, thus identifying the market potential of mobile Internet information distribution in the situation that the domestic Internet industry has been overwhelmingly divided up by BAT. The excellent traits possessed by entrepreneurs are the prerequisite for them to identify entrepreneurial opportunities and develop them, which are necessary to carry out entrepreneurial activities.

TABLE 1 Analysis of Byte companies by entrepreneurs

Secondary coding	Primary coding	Examples of entrepreneurs
Social traits	Technical knowledge	Has worked as a software engineer and has accumulated rich technical knowledge.
	Entrepreneurial experience	Developed a system office system that failed; he also started an independent business in a mobile Internet environment and achieved initial success.
	Personal network	Half of the engineers in charge of technology in the company's initial team were Zhang Yiming's former colleagues.
Individual traits	Alertness	Perceiving the mobile Internet trend, found a lot of space for the organisation and distribution of information.
	Innovativeness	Passionate about learning new things and especially thoughtful.
	Adventurousness	Uncomfortable with routine, willing to take risks, and embrace uncertainty.
	Achievement needs	Not satisfied with his current million-dollar-a-year job, he wanted to achieve a bigger dream.

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4.2 | Creation path of ByteDance from the perspective of opportunity

ByteDance uses AI to construct the overall underlying logic of the product, AI algorithm technology is used to support the information matching efficiency, and the main product business ecology is created during the start-up process. The company provides personalised information service for all users through "AI + content" and thus gathers a large amount of data traffic, creates more data flow for the company by exploring other multiple content consumption scenarios, and then promotes the efficient conversion of data flow into value. In this way, the company creates traffic and realises cash flow at the same time, forming a perfect "traffic-value" business ecological closed loop. Correspondingly, from the perspective of opportunity, the process of creating the company's main business goes through three stages: "cultivating opportunity soil, building entrepreneurial opportunity, and feeding opportunity soil". In this study, the entrepreneurial paths of ByteDance are coded (Table 2), and the development paths of entrepreneurship of such companies are summarised as follows.

First, Byte has initially completed the construction of the company's traffic through building traffic entrance, attracting original creators, model innovation, traffic synergy, and AI technology development. On the one hand, the company provides user traffic entrance by developing business products, attracts excellent original authors, and designs a new business model "algorithm + content" based on AI technology to achieve efficient distribution of information and to create a large amount of traffic for the company. Using the data traffic advantages that have accumulated, the company has started the layout of the short video field, realised the traffic synergy between different platforms such as Toutiao and Douyin, and thus gradually break through the entrepreneurial boundary between different fields. On the other hand, the company pays full attention to the powerful role of AI technology. The company weighs and addresses the impact of AI technology on entrepreneurship. For example, the company attaches great importance to the impact of intelligent recommendation on original authors, seeks for a way to balance the contradiction between "algorithm recommendation" and "active attention", and creates a win-win situation of efficient information distribution, satisfying users' personalised needs and attracting more original authors. On this basis, the company continues to attract and deposit user flow steadily. In addition, AI is also an

TABLE 2 Creation paths for ByteDance from the perspective of opportunity

Opportunity view	Secondary coding	Primary coding	Typical examples of ByteDance
Cultivating opportunity soil	Data traffic building	Data traffic entrance	To develop platforms such as Toutiao and Douyin to meet users' demands for information they care about and video content they are interested in.
		Attracting original creators	Focus on attracting high-quality original authors, balance the information flow between "algorithm recommendation" and "active attention", and give creators space to manage fans.
		Model innovation	AI technology is used to change the traditional way of information distribution, and "algorithm + content" mode is used to make personalised recommendations to catch users' needs.
		Flow synergy	The general information platform with Toutiao as the core and the short video platform with Douyin as the core cooperate and develop together.
		AI Technology R&D	AI lab has been established, committed to machine learning, natural language understanding, computer vision, data mining and other directions, and aimed at the long-term and open problems in the field of AI to provide a good support for the company's future industrial development.

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TABLE 2 (Continued)

Opportunity view	Secondary coding	Primary coding	Typical examples of ByteDance
Building opportunities	Data traffic expansion	Deployment of overseas markets	Relying on AI technology, replicate domestic products from 0 to 1.
		Grasp the market segment	The neglected “"To B” market also has a huge dividend that has not been fully exploited. ByteDance developed the office suite Lark To grab the domestic and foreign “To B” market.
		Mining vertical fields	ByteDance taps into existing traffic by laying out its products in education, social, finance, reading, and more.
		Create a content incubator	The incubator has been established in 10 cities, including Beijing, and has incubated more than 200 content start-ups.
		Create content ecology	ByteDance needs to continuously produce quality content to maintain its traffic growth and accumulation, so invest in various content production organisations that complement Byte's products.
Feeding opportunity soil	Data traffic monetisation	Develop e-commerce business	Established the “E-commerce” business department, coordinated multiple content platforms such as Toutiao, Douyin, and Watermelon Video, combined traffic with commodity trading, and supplemented the company's shortcomings in traffic terminals.
		Tapping the potential of data traffic	To take full advantage of the potential of the retained user traffic data, a “local direct business center” has been set up to explore customers in the life services, culture, tourism, and catering industries.
		Increase user engagement	In the Internet scenario, the growth of user time tends to stagnate, and the user time of ByteDance products keeps increasing year on year. The average daily length of the three ByteDance products is all over 60 min.
		Efficient traffic conversion	The global total MAU of the products exceeded 1.5 billion, and the company's traffic was effectively transformed. The market share of online advertising increased rapidly, accounting for more than 87% of the company's total revenue.

important support technology for developing business ventures. The empowerment of AI technology makes the company's content distribution more intelligent, personalised, socialised, and universal, helping the company create explosive products quickly and becoming the core technology that the company relies on for user traffic gathering and rapid business development.

Moreover, AI technological progress requires a large amount of data and continuous interaction with users to learn, so AI progress and the company's growth complement each other. Therefore, the company created AI Lab to devote itself to AI technology research and development and focus on advanced technologies related to AI, such as VR and AR, which will help the company firmly grasp the opportunities for future development of the Internet industry. Therefore, the data flow built by the company based on AI technology and the continuous promotion of AI research and development are the “soil”

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for the company to expand other business opportunities in the future.

Moreover, the company has built the company's diversified business and expanded more content-related business scenarios by laying out overseas markets, grasping niche markets, building vertical businesses, creating content incubators, and building content ecological measures, which help company break through the dilemma of topping out Internet traffic and slowing down the company's traffic growth and provide the company with a source of increased traffic. With the development and progress of AI technology, content-based application scenarios will also become richer and richer, and user demand has endless potential, which is a huge opportunity for ByteDance to grow. Therefore, AI technology can greatly expand the new scenarios of content demand. ByteDance builds entrepreneurial opportunities applied to different scenarios, innovatively provides high-quality content products for these scenarios, and gradually builds out its own diversified product ecology. The perfect ecology ensures the large-scale application of AI technology. AI technology can also influence the perfection of corporate content ecology and realise further traffic expansion of the company. Unlike traditional business management requirements, user traffic is the key to the survival and development of Internet-type companies, and traffic growth is critical to Internet companies. Therefore, the phase in which the company expands the multiple scenarios of content consumption to increase traffic is the process in which the company builds more entrepreneurial opportunities to achieve growth.

Finally, the company gradually develops live e-commerce business, develops local business to tap the traffic potential, and improves corporate ecology through efficient traffic conversion from advertising business. It increases the user's length of use to enhance user stickiness, solidify the company's traffic pool, and realise the value of the company's massive traffic. The company's AI technology empowerment has also increased the effectiveness of traffic monetisation. On the one hand, the content ecology built by ByteDance provides a perfect scenario for AI technology to be implemented. On the other hand, the company uses AI to analyse text, images, and videos to obtain a large amount of metadata and uses AI to obtain a "complete portrait" of users and then match them with their specific needs to ensure the accurate placement of advertisements. In this way, the company can improve the ROI of advertisements and GMV of product transactions. At this stage, the company can realise the transformation of "traffic-business value" and form a gradually improved enterprise growth ecology. The value and traffic accumulated by the company will become a solid foundation for the company to build data traffic further and create entrepreneurial opportunities, thus feeding the "soil" for the company to cultivate opportunities.

The company's entrepreneurial growth revolves around the three levels of "traffic construction, traffic expansion, and traffic monetisation", forming a closed loop of enterprise ecology with traffic as the core (Figure 2). The company takes the lead in relying on "AI + big data" technology to build the corporate traffic of information and short video, which has accumulated a

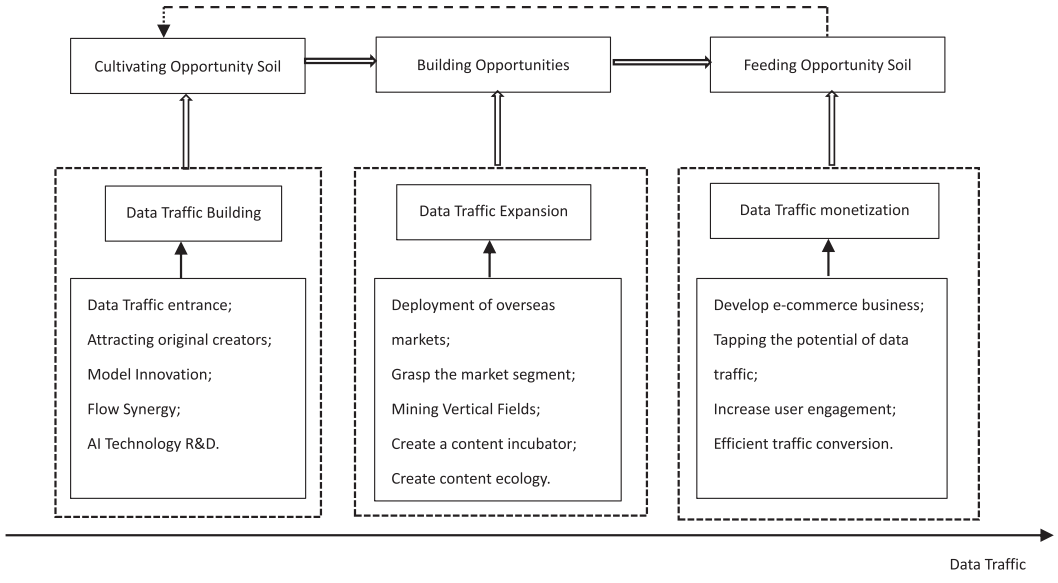


FIGURE 2 Creation paths for ByteDance from the perspective of opportunity

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large amount of data traffic for the company. In addition, based on the characteristics of large traffic, which are reusable, extensible, easy to update and distributed without boundaries (Saint & Walker, 2014), and relatively mature AI technology support, it is helpful for the company to build new entrepreneurial opportunities in other market segments and build a multi-content product matrix to improve the content ecology and to maintain the company's user traffic continued to expand.

Ultimately, the accurate matching of AI technology will improve the delivery efficiency of the company's advertising business and the commodity transaction rate of e-commerce business, and other ways to effectively convert the company's massive traffic into commercial value. Through these three stages, a relatively complete closed-loop ecological path between "flow value" of the enterprise is formed to improve the enterprise's business efficiency and promote its rapid growth.

4.3 | The entrepreneurial process of ByteDance company

After coding and analyzing the information, this study finally summarised the entrepreneurial process of ByteDance (Figure 3). Entrepreneurs with extensive entrepreneurial knowledge and full of innovation, risk-taking, alertness, and the need for achievement are able to seize entrepreneurial opportunities (Zhong & Lu, 2014). In perceiving the potential of Internet changes and AI technology advancement and application, entrepreneurs find great room for improvement in the organisation and distribution of online information and rely on AI technology for data traffic construction, expansion, and monetisation. Finally, the entrepreneur creates the business growth ecology of the enterprise. The scalability and continuous regeneration of traffic in the Internet industry have weakened the boundaries of entrepreneurship, promoting the opportunity for AI technology and products and services to reconfigure and create new scenarios and markets. From the perspective of opportunity, the company creates a complete entrepreneurial ecology for new enterprises by "cultivating opportunity soil, building entrepreneurial opportunity, and feeding opportunity soil" to realise the creation of new value. With the development and growth of new enterprises, the impact of new technologies such as virtual reality, scientific computing, and other technologies on human life gradually emerges. The competition in the Internet industry intensifies. Both entrepreneurs and companies are facing more and more challenges, which require entrepreneurs to continuously expand their thinking and think about the next stage of growth strategies and opportunities. Therefore, at this stage, entrepreneurs adjust their management focus to focus on the company's vision and strategy, corporate culture, social responsibility, and other important things for the company's long-term development and eventually promote the company's development and growth in an upward spiral manner.

5 | DISCUSSION

5.1 | Conclusion of the study

This study takes ByteDance as a case study to analyse the entrepreneurial process of AI-enabled start-ups in the Internet industry and contributes to the existing AI entrepreneurship research. The entrepreneurs successfully use the technology empowerment of "AI + Big Data" to lead ByteDance to grow rapidly within a short period of time after its establishment. From the perspective of

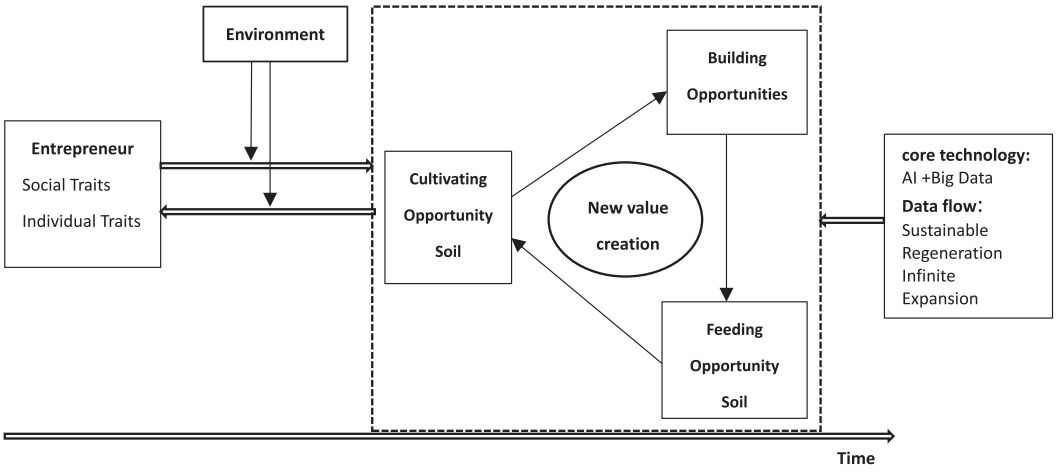


FIGURE 3 ByteDance's entrepreneurial process model

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opportunity, an entrepreneur with rich a priori knowledge and unique personal characteristics can keenly identify the potential of AI technology and big data in the Internet industry and is the first to seize the entrepreneurial opportunity of the AI-enabled Internet industry. Relying on AI technology, entrepreneurs use "algorithm + content" to create "traffic + data" to form the core competitiveness of enterprises, which create new entrances, new models, and new values of the Internet. The entrepreneurs build, expand, and monetise the flow, hence build a complete "flow value" enterprise ecology. Accordingly, entrepreneurs have completed the creation and growth of new enterprises by cultivating opportunity soil, building new opportunities, and feeding opportunity soil. Entrepreneurs are not satisfied with the success of their business at present. With the development and progress of 5G technology, general AI technology, the Internet of Things, quantum computing, and other new technologies (Golpîra et al., 2021; Li et al., 2018; Sigov et al., 2022; Viriyasitavat et al., 2019; Xu, 2011; Xu & Viriyasitavat, 2019), the competition in the Internet industry will become more and more fierce. Entrepreneurs will design long-term goals and strategies for the long-term development of their companies so as to lead them to further growth.

5.2 | Practice insights

In the era of digital intelligence, relying on Enabling Technology of AI, the Internet, health care, finance, education, robotics, and other fields will generate a large number of entrepreneurial opportunities. AI is the main technology to improve social productivity and solve the imbalance between supply and demand in the future and will become an important engine for the development of various industries. The successful use of AI technology affects start-ups' competitive advantage and growth, especially in the rapidly changing Internet industry. The rapid rise of ByteDance is an outstanding representative of the Internet industry benefiting from AI technology, and it also provides several realistic inspirations for Internet industry start-ups.

First of all, grasping the real needs of users: With the development of the Internet, users' needs are becoming more personalised, intelligent, and scenario-oriented, and the Internet industry will face more detailed and fierce competition. ByteDance captures the time and scene requirements of users' fragmented consumption and recommends personalised information for users, which is welcomed by a large number of users. However, the personalised needs of users are still the blue ocean of entrepreneurship in the Internet industry. Whether the personalised needs of all users can be met will directly affect the success of entrepreneurial enterprises.

Second, taping the potential of AI technology: Technological change is the most important external factor affecting entrepreneurial activities. At present, the development of emerging technologies such as big data, the Internet of Things, and AI has created many opportunities for entrepreneurial activities. However, only by truly fully tapping the potential of new technologies can it provide a strong impetus for entrepreneurship. AI technology has brought a new and stronger growth force to the Internet industry (Broo et al., 2022; Zhang & Lu, 2021). AI technology empowerment brings new and stronger growth forces to the Internet industry. With the research and development of "5G + AI" and general AI, more new technologies, new products, new models, and new forms of business will be born, which will bring disruptive innovation to all industries, change the social and economic development and the way people work and live, and bring more opportunities and challenges to entrepreneurial activities.

In addition, taking advantage of the changing entrepreneurial environment: Excellent entrepreneurs are good at taking advantage of accidental factors in the entrepreneurial environment to explore potential entrepreneurial opportunities. Just as Zhang Yiming, the founder of ByteDance, insists on "embracing uncertainty". The only constant in today's era is change. The innovation of new technology is getting faster and faster. Social production and lifestyle are constantly changing. Consumers' habits are also changing constantly. Only by embracing these changes and following the trend of The Times can entrepreneurs seize the opportunity to start a business.

Finally, maintaining innovation: For start-ups, innovation includes technological, management, and business model innovation. Internet innovation has no boundary. The subversive innovation of ByteDance has broken through the traditional entrepreneurial model. However, with the progress of AI and other advanced technologies, the Internet industry still has huge development potential.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable. No datasets were generated during the current study.

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REFERENCES

Agrawal, A., Gans, J. S., & Goldfarb, A. (2017). What to expect from AI. *MIT Sloan Management Review*, *58*(3), 23–26.

10991743a, 2022, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/sres.2867 by Blekinge Tekniska Hogskola, Wiley Online Library on [09/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Alvarez, S., & Barney, J. B. (2007). Discovery and creation: Alternative theories of entrepreneurial action. *Social Science Electronic Publishing*, *1*(1–2), 11–26. https://doi.org/10.2139/ssrn.900200

Ao, J., & Liu, Z. (2014). What impact entrepreneurial intention? Cultural, environmental, and educational factors. *Journal of Management Analytics*, *1*(3), 224–239. https://doi.org/10.1080/23270012.2014.994232

Bansal, G., Anand, A., & Aggrawal, D. (2021). Modeling multi-generational diffusion for competitive brands: An analysis for telecommunication industries. *Journal of Management Analytics*, *8*(4), 715–740. https://doi.org/10.1080/23270012.2021.1881925

Broo, D. G., Kaynak, O., & Sait, S. M. (2022). Rethinking engineering education at the age of industry 5.0. *Journal of Industrial Information Integration*, *25*, 100311. https://doi.org/10.1016/j.jii.2021.100311

Brown, T. E. (2017). Sensor-based entrepreneurship: A framework for developing new products and services. *Business Horizons*, *60*(6), 819–830. https://doi.org/10.1016/j.bushor.2017.07.008

Bruyat, C., & Julien, P. A. (2001). Defining the field of research in entrepreneurship. *Journal of Business Venturing*, *16*(2), 165–180. https://doi.org/10.1016/S0883-9026(99)00043-9

Cai, L., Yang, Y., & Lu, S. (2019). Review and Prospect of the research on the impact of digital technology on entrepreneurial activities. *Studies in Science of Science*, *37*(10), 1816–1824.

Chen, H., & Lei, J. (2021). Research and development of "science-based Entreprenenurial firms" and industrial transformation mechanism: Case study approach. *Journal of Industrial Integration and Management*, *6*(04), 383–405. https://doi.org/10.1142/S2424862220500153

Cockburn, I. M., Henderson, R., & Stern, S. (2018). The impact of AI on innovation. Nber Chapters. https://doi.org/10.3386/w24449

Davidsson, P., Recker, J., & Briel, F. V. (2017). External enablers in new venture creation processes: A framework. *Academy of Management Perspectives*, *34*(3), 311–332. https://doi.org/10.5465/AMBPP.2017.12

Du, C. Z., & Liu, Z. P. (2019). Innovation mechanism and policy incentive effect of academic entrepreneurship-numerical simulation analysis based on data of A-share listed companies in AI industry. *Research on Economics and Management*, *40*(6), 119–130. https://doi.org/10.13502/j.cnki.issn1000-7636.2019.06.009

Dutta, D. K., & Crossan, M. M. (2010). The nature of entrepreneurial opportunities: Understanding the process using the 4i organizational learning framework. *Entrepreneurship: Theory and Practice*, *29*(4), 425–449. https://doi.org/10.1111/j.1540-6520.2005.00092.x

Eckhardt, J. T., & Shane, S. A. (2003). Opportunities and entrepreneurship. *Journal of Management*, *29*(3), 333–349. https://doi.org/10.1016/S0149-2063(02)00225-8

Golpîra, H., Khan, S. A. R., & Safaeipour, S. (2021). A review of logistics internet-of-things: Current trends and scope for future research. *Journal of Industrial Information Integration*, *22*, 100194. https://doi.org/10.1016/j.jii.2020.100194

Hou, F., Ge, B., & Su, Y. (2018). Review of classical intention-based entrepreneurial models and construction of framework from combined perspectives of entrepreneurial motivation and cognition. *Journal of Industrial Integration and Management*, *3*(03), 1850012. https://doi.org/10.1142/S2424862218500124

Khalid, N. (2020). AI learning and entrepreneurial performance among university students: Evidence from Malaysian higher educational institutions. *Journal of Intelligent Fuzzy Systems*, *39*(4), 5417–5435. https://doi.org/10.3233/JIFS-189026

Khan, W., Ali, T., & Dhamija, A. (2022). Perceived obstacles and performance of food and agribusiness enterprises: Implications for urban and rural entrepreneurship development. *Journal of Industrial Integration and Management*, 1–18. https://doi.org/10.1142/S2424862221500287

Li, S., Xu, L., & Zhao, S. (2018). 5G internet of things: A survey. *Journal of Industrial Information Integration*, *10*, 1–9. https://doi.org/10.1016/j.jii.2018.01.005

Li, nY., Shan, BB., Fei, YP., & Li BW. (2021). Digital technology entrepreneurship: Review and prospect of research topics. *R and D Management*, *33*(1), 65–77. https://doi.org/10.13581/j.cnki.rdm.20190926

Lin, S., & Jiang, Y. F. (2005). Review and analysis of research progress in entrepreneurship. *Modernization of Management*, *3*(6), 22–24.

Liu, S. S., Song, L. F., & Song, Y. F. (2020). Corporate entrepreneurial path of internet unicorn enterprises in digital context. *Studies in Science of Science*, *38*(01), 116–126.

Liu, Z., Ding, F., Xiao, Y. Z., & Cui, G. L. (2019). Research on the mechanism of opportunity recognition and development of social entrepreneurship based on the theory of resource bricolage. *Chinese Journal of Management*, *016*(007), 1006–1015.

Liu, Z. Y., & Wang, Z. M. (2020). AI enabling entrepreneurship: Comparison of theoretical frameworks. *Foreign Economies & Management*, *42*(12), 3–16. https://doi.org/10.16538/j.cnki.fem.20201020.101

Luo, Y., & Bai, Y. (2021). Business model innovation of technical start-ups in emerging markets. *Journal of Industrial Integration and Management*, *6*(03), 319–332. https://doi.org/10.1142/S2424862221500202

Politis, D. (2010). The process of entrepreneurial learning: A conceptual framework. *Entrepreneurship Theory & Practice Journal*, *29*(4), 399–424. https://doi.org/10.1111/j.1540-6520.2005.00091.x

Sahlman, W. A. (1999). *Some Thoughts on Business Plans*. Harvard Business School Publication.

Saint, J., & Walker. (2014). Big data: A revolution that will transform how we live, work, and think. *International Journal of Advertising: The Review of Marketing Communications*, *33*(1), 181–183. https://doi.org/10.2501/IJA-33-1-181-183

Shane, S., & Venkataraman, S. (2000). The promise of entrepreneurship as a field of research. *Academy of Management Review*, *25*(1), 217–226. https://doi.org/10.1007/978-3-540-48543-8_8

Si, X. F., Wang, S., & Fu, Y. (2016). Where do entrepreneurial opportunities come from: Discover, build or discover + build?—Theoretical frontier research on entrepreneurial opportunities. *Management World*, *270*(3), 115–127.

Sigov, A., Ratkin, L., Ivanov, L., & Xu, L. (2022). Emerging enabling technologies for industry 4.0 and beyond. *Information Systems Frontiers*, 1–11. https://doi.org/10.1007/s10796-021-10213-w

10991743a, 2022, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/sres.2867 by Blekinge Tekniska Hogskola, Wiley Online Library on [09/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Timmons, J., & Spinelli, A. Jr. (2004). *New venture creation* (6th ed.). McGraw-Hill Company Inc.

Vaghely, I. P., & Venkataraman, S. (2010). Are opportunities recognized or constructed?: An information perspective on entrepreneurial opportunity identification. *Journal of Business Venturing*, *25*(1), 73–86. https://doi.org/10.1016/j.jbusvent.2008.06.004

Viriyasitavat, W., Anuphaptrirong, T., & Hoonsopon, D. (2019). When blockchain meets internet of things: Characteristics, challenges, and business opportunities. *Journal of Industrial Information Integration*, *15*, 21–28. https://doi.org/10.1016/j.jii.2019.05.002

Wang, Q., & Cai, L. (2011). Study on the process and influencing factors of entrepreneurial opportunity development. *Study & Exploration*, (03), 191–193. https://doi.org/10.3969/j.issn.1002-462X.2011.03.047

Wang, X. F. (2016). A literature review on entrepreneur's behaviors: Connecting entrepreneur's individual factors, entrepreneurial process and outcomes. *Science of Science and Management of S. & T*, *37*(8), 3–19.

Wu, H. (2016). AI: The next startup. *Money*, *2*(7), 23–24. https://doi.org/10.3969/j.issn.1673-1107.2016.07.013

Xu, L. (2011). Enterprise systems: State-of-the-art and future trends. *IEEE Transactions on Industrial Informatics*, *7*(4), 630–640. https://doi.org/10.1109/TII.2011.2167156

Xu, L., & Viriyasitavat, W. (2019). Application of blockchain in collaborative internet-of-things services. *IEEE Transactions on Computational Social Systems*, *6*(6), 1295–1305. https://doi.org/10.1109/TCSS.2019.2913165

Xue, J. (2018). A literature review on entrepreneur's traits, social capital and entrepreneurial enterprises performance. *Modernization of Management*, *38*(06), 128–131. https://doi.org/10.19634/j.cnki.11-1403/c.2018.06.030

Yeşilkaya, M., & Yıldız, T. (2021). Optimistic expectations and proactivity: Can they be the key to successful strategies in entrepreneurship? *Journal of Industrial Integration and Management*, *07*, 1–20. https://doi.org/10.1142/S2424862221500238

Yu, J., Meng, Q. S., Zhang, Y., & Jin, J. (2018). Digital entrepreneurship: The future directions of entrepreneurship theory and practice in the digital era. *Studies in Science of Science*, *036*(010), 1801–1808.

Zhang, C., & Lu, Y. (2021). Study on artificial intelligence: The state of the art and future prospects. *Journal of Industrial Information Integration*, *23*, 100224. https://doi.org/10.1016/j.jii.2021.100224

Zhang, Y. (2018). *An Overview of Classic Entrepreneurship Literature*. Nankai University Press.

Zheng, Q. (2017). A brief analysis of content entrepreneurship opportunities in the early stage of AI instrumentation. *Science-Technology and Publication*, *4*(11), 11–14.

Zhong, W. R., & Lu, C. R. (2014). Study on the influence path from environmental dynamism to feasibility of entrepreneurial opportunity identification-centered on entrepreneurial traits. *Forecast*, *33*(3), 27–33.

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10991743a, 2022, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/sres.2867 by Blekinge Tekniska Hogskola, Wiley Online Library on [09/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License